

CSE 460: SOFTWARE ANALYSIS AND DESIGN

Homework #2 — Heart Health Imaging and Recording System

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Item 1 Working Code

The complete implementation of the Heart Health Imaging and Recording System is written in Java with a JavaFX GUI. The source code is hosted on GitHub in *HeartHealthImagingSystem* folder. The repository contains the full project including all source files, the build configuration, and sample data files demonstrating each of the four functional areas:

- **Patient Intake** — register new patients and schedule CT scan appointments
- **CT Scan Technician** — enter and save Agatston CAC score results
- **Patient View** — patients log in with their 5-digit ID to view CT results
- **Doctor View** — heart specialist reviews results, determines risk, and emails patient

GitHub repository link (code):

<https://github.com/jumperpc3000/cse460-software-analysis-design/tree/main/Assignments/HW2>

Item 2 Demo Video

A video demonstration is provided showing the full system in action. The demo covers:

- a. How to run the project and login
- b. Patient Intake functionality — entering patient data and scheduling a scan
- c. CT Scan Tech functionality — recording Agatston CAC scores
- d. Patient View functionality — logging in by patient ID and viewing results
- e. Doctor View functionality — reviewing results, risk assessment, and sending the patient email

Demo video link (Zoom, Passcode: m²Wn1GG):

https://asu.zoom.us/rec/share/48etXgGKyy4NSEb8YIXB7PvD_p1x8tp2v8Xlewhde41QS3pGlq2wn0oVtEJY1uY1.NTFs2VwHw2kWu_EXPasscode:m²Wn1GG

Item 3 Final Implementation Class Diagram

The final class diagram reflecting the completed implementation is presented on the following page. It was produced in **Astah** and captures all packages, classes, fields, methods, and relationships of the delivered system.

CSE 460 Website (Assignments):

<https://jumperpc3000.github.io/cse460-software-analysis-design/#assignments>



